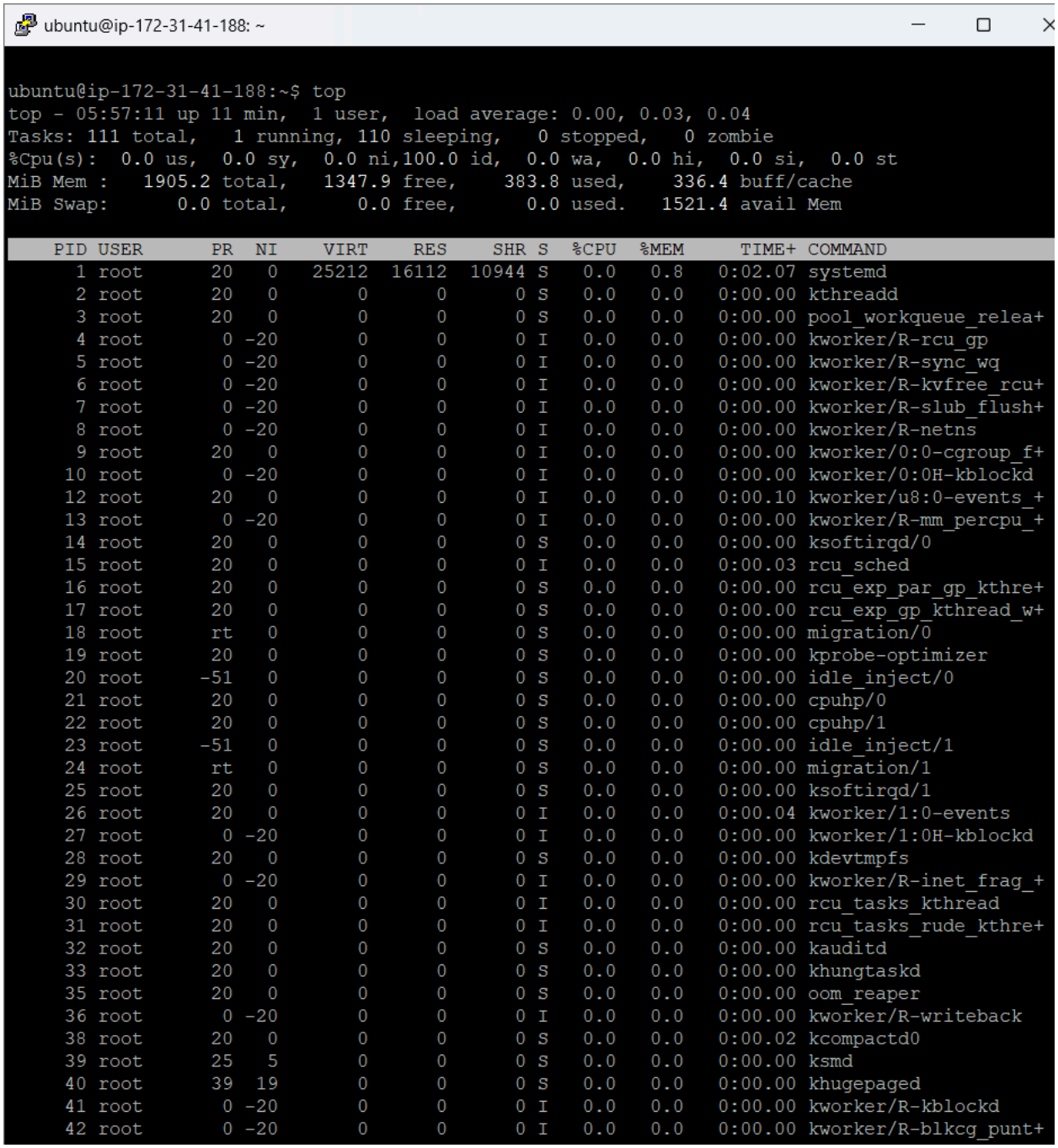
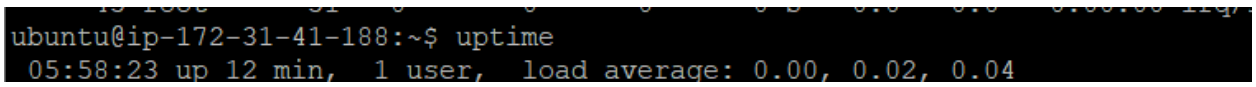
CPU and load


ubuntu@ip-172-31-41-188: ~
ubuntu@ip-172-31-41-188:~$ top
top - 05:57:11 up 11 min, 1 user, load average: 0.00, 0.03, 0.04
Tasks: 111 total,   1 running, 110 sleeping,   0 stopped,   0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem :   1905.2 total,   1347.9 free,   383.8 used,   336.4 buff/cache
MiB Swap: 0.0 total,   0.0 free,   0.0 used.   1521.4 avail Mem
PID USER PR NI VIRT RES SHR S %CPU %MEM TIME+ COMMAND
1 root 20 0 25212 16112 10944 S 0.0 0.8 0:02.07 systemd
2 root 20 0 0 0 0 S 0.0 0.0 0:00.00 kthreadd
3 root 20 0 0 0 0 S 0.0 0.0 0:00.00 pool_workqueue_relea+
4 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-rcu_gp
5 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-sync_wq
6 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-kvfree_rcu+
7 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-slub_flush+
8 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-netns
9 root 20 0 0 0 0 I 0.0 0.0 0:00.00 kworker/0:0-cgroup_f+
10 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/0:0H-kblockd
12 root 20 0 0 0 0 I 0.0 0.0 0:00.10 kworker/u8:0-events_+
13 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-mm_percpu_+
14 root 20 0 0 0 0 S 0.0 0.0 0:00.00 ksoftirqd/0
15 root 20 0 0 0 0 I 0.0 0.0 0:00.03 rcu_sched
16 root 20 0 0 0 0 S 0.0 0.0 0:00.00 rcu_exp_par_gp_kthre+
17 root 20 0 0 0 0 S 0.0 0.0 0:00.00 rcu_exp_gp_kthread_w+
18 root rt 0 0 0 0 S 0.0 0.0 0:00.00 migration/0
19 root 20 0 0 0 0 S 0.0 0.0 0:00.00 kprobe-optimizer
20 root -51 0 0 0 0 S 0.0 0.0 0:00.00 idle_inject/0
21 root 20 0 0 0 0 S 0.0 0.0 0:00.00 cpuhp/0
22 root 20 0 0 0 0 S 0.0 0.0 0:00.00 cpuhp/1
23 root -51 0 0 0 0 S 0.0 0.0 0:00.00 idle_inject/1
24 root rt 0 0 0 0 S 0.0 0.0 0:00.00 migration/1
25 root 20 0 0 0 0 S 0.0 0.0 0:00.00 ksoftirqd/1
26 root 20 0 0 0 0 I 0.0 0.0 0:00.04 kworker/1:0-events
27 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/1:0H-kblockd
28 root 20 0 0 0 0 S 0.0 0.0 0:00.00 kdevtmpfs
29 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-inet_frag_+
30 root 20 0 0 0 0 I 0.0 0.0 0:00.00 rcu_tasks_kthread
31 root 20 0 0 0 0 I 0.0 0.0 0:00.00 rcu_tasks_rude_kthre+
32 root 20 0 0 0 0 S 0.0 0.0 0:00.00 kauditd
33 root 20 0 0 0 0 S 0.0 0.0 0:00.00 khungtaskd
35 root 20 0 0 0 0 S 0.0 0.0 0:00.00 oom_reaper
36 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-writeback
38 root 20 0 0 0 0 S 0.0 0.0 0:00.02 kcompactd0
39 root 25 5 0 0 0 S 0.0 0.0 0:00.00 ksmd
40 root 39 19 0 0 0 S 0.0 0.0 0:00.00 khugepaged
41 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-kblockd
42 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-blkcg_punt+


ubuntu@ip-172-31-41-188:~$ uptime
05:58:23 up 12 min, 1 user, load average: 0.00, 0.02, 0.04

```
ubuntu@ip-172-31-41-188:~$ nproc
2
ubuntu@ip-172-31-41-188:~$ mpstat -P ALL 1 5
Linux 7.0.0-1006-aws (ip-172-31-41-188)         10/20/26        _x86_64_        (2 CPU)

05:59:01     CPU    %usr   %nice    %sys %iowait    %irq   %soft  %steal  %guest  %gnice   %i
dle
05:59:02     all    0.00    0.00    0.00    0.00    0.00    0.00    0.00    0.00    0.00  100
.00
05:59:02       0    0.00    0.00    0.00    0.00    0.00    0.00    0.00    0.00    0.00  100
.00
05:59:02       1    0.00    0.00    0.00    0.00    0.00    0.00    0.00    0.00    0.00  100
.00

05:59:02     CPU    %usr   %nice    %sys %iowait    %irq   %soft  %steal  %guest  %gnice   %i
dle
05:59:03     all    0.00    0.00    0.00    0.00    0.00    0.00    0.00    0.00    0.00  100
.00
05:59:03       0    0.00    0.00    0.00    0.00    0.00    0.00    0.00    0.00    0.00  100
.00
05:59:03       1    0.00    0.00    0.00    0.00    0.00    0.00    0.00    0.00    0.00  100
.00

05:59:03     CPU    %usr   %nice    %sys %iowait    %irq   %soft  %steal  %guest  %gnice   %i
dle
05:59:04     all    0.00    0.00    0.00    0.00    0.00    0.00    0.00    0.00    0.00  100
.00
05:59:04       0    0.00    0.00    0.00    0.00    0.00    0.00    0.00    0.00    0.00  100
.00
05:59:04       1    0.00    0.00    0.00    0.00    0.00    0.00    0.00    0.00    0.00  100
.00

05:59:04     CPU    %usr   %nice    %sys %iowait    %irq   %soft  %steal  %guest  %gnice   %i
dle
05:59:05     all    0.50    0.00    0.00    0.00    0.00    0.00    0.00    0.00    0.00   99
```

```
05:59:03     CPU    %usr   %nice    %sys %iowait    %irq   %soft  %steal  %guest  %gnice   %i
dle
05:59:04     all    0.00    0.00    0.00    0.00    0.00    0.00    0.00    0.00    0.00  100
.00
05:59:04       0    0.00    0.00    0.00    0.00    0.00    0.00    0.00    0.00    0.00  100
.00
05:59:04       1    0.00    0.00    0.00    0.00    0.00    0.00    0.00    0.00    0.00  100
.00

05:59:04     CPU    %usr   %nice    %sys %iowait    %irq   %soft  %steal  %guest  %gnice   %i
dle
05:59:05     all    0.50    0.00    0.00    0.00    0.00    0.00    0.00    0.00    0.00   99
.50
05:59:05       0    1.00    0.00    0.00    0.00    0.00    0.00    0.00    0.00    0.00   99
.00
05:59:05       1    0.00    0.00    0.00    0.00    0.00    0.00    0.00    0.00    0.00  100
.00

05:59:05     CPU    %usr   %nice    %sys %iowait    %irq   %soft  %steal  %guest  %gnice   %i
dle
05:59:06     all    0.00    0.00    0.50    0.00    0.00    0.00    0.00    0.00    0.00   99
.50
05:59:06       0    0.00    0.00    0.99    0.00    0.00    0.00    0.00    0.00    0.00   99
.01
05:59:06       1    0.00    0.00    0.00    0.00    0.00    0.00    0.00    0.00    0.00  100
.00

Average:     CPU    %usr   %nice    %sys %iowait    %irq   %soft  %steal  %guest  %gnice   %i
dle
Average:     all    0.10    0.00    0.10    0.00    0.00    0.00    0.00    0.00    0.00   99
.80
Average:       0    0.20    0.00    0.20    0.00    0.00    0.00    0.00    0.00    0.00   99
.60
Average:       1    0.00    0.00    0.00    0.00    0.00    0.00    0.00    0.00    0.00  100
.00
```

```
ubuntu@ip-172-31-41-188:~$ vmstat 1 5
procs -----------memory---------- ---swap-- -----io---- -system-- -------cpu-------
 r  b   swpd   free   buff  cache   si   so    bi    bo   in   cs us sy id wa st gu
 0  0      0 1381280  17532 327716    0    0   311   750  103    0  1  0 99  0  0  0
 0  0      0 1381280  17532 327716    0    0     0     0   83   59  0  0 100  0  0  0
 0  0      0 1381232  17532 327772    0    0     0     0   53   43  0  0 100  0  0  0
 0  0      0 1381232  17532 327772    0    0     0     0   54   43  0  0 100  0  0  0
 0  0      0 1381232  17532 327772    0    0     0     0   47   36  0  0 100  0  0  0
ubuntu@ip-172-31-41-188:~$ ps aux --sort=-%cpu | head -10
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.1  0.8  25212 16112 ?        Ss   05:45   0:02 /sbin/init
root        1133  0.0  1.0 1759024 21008 ?       Ssl  05:45   0:00 /snap/amazon-ssm-agent/130
09/amazon-ssm-agent
root         736  0.0  2.0 2025016 40292 ?       Ssl  05:45   0:00 /usr/lib/snapd/snapd
root         203  0.0  0.6  39048 13648 ?        Ss   05:45   0:00 /usr/lib/systemd/systemd-u
devd
root         129  0.0  0.7  50116 14108 ?        S<s  05:45   0:00 /usr/lib/systemd/systemd-j
ournald
ubuntu      1220  0.0  0.6  21880 12932 ?        Ss   05:55   0:00 /usr/lib/systemd/systemd -
-user
root         729  0.0  1.5  46808 29892 ?        Ss   05:45   0:00 /usr/bin/python3 /usr/bin/
networkd-dispatcher --run-startup-triggers
root         848  0.0  1.6 126208 32260 ?        Ssl  05:45   0:00 /usr/bin/python3 /usr/shar
e/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
polkitd      732  0.0  0.5 381772 10216 ?        Ssl  05:45   0:00 /usr/lib/polkit-1/polkitd
--no-debug --log-level=notice
ubuntu@ip-172-31-41-188:~$
```

Memory and swap

```
ubuntu@ip-172-31-41-188:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           1.9Gi       382Mi       1.3Gi       2.7Mi       337Mi       1.5Gi
Swap:             0B          0B          0B
ubuntu@ip-172-31-41-188:~$ cat /proc/meminfo
MemTotal:        1950940 kB
MemFree:         1378232 kB
MemAvailable:    1557052 kB
Buffers:           17548 kB
Cached:           306880 kB
SwapCached:            0 kB
Active:           304024 kB
Inactive:          92672 kB
Active(anon):      96968 kB
Inactive(anon):        0 kB
Active(file):     207056 kB
Inactive(file):    92672 kB
Unevictable:       25952 kB
Mlocked:            9320 kB
SwapTotal:             0 kB
SwapFree:              0 kB
Zswap:                 0 kB
Zswapped:              0 kB
Dirty:                 0 kB
Writeback:             0 kB
AnonPages:         98304 kB
Mapped:           103056 kB
Shmem:              2776 kB
KReclaimable:      21324 kB
Slab:              67928 kB
SReclaimable:      21324 kB
SUnreclaim:        46604 kB
KernelStack:        2456 kB
PageTables:         3756 kB
SecPageTables:         0 kB
NFS_Unstable:          0 kB
Bounce:                0 kB
WritebackTmp:          0 kB
CommitLimit:      975468 kB
Committed_AS:     577516 kB
VmallocTotal:   34359738367 kB
VmallocUsed:       13404 kB
VmallocChunk:          0 kB
Percpu:             1064 kB
HardwareCorrupted:     0 kB
AnonHugePages:         0 kB
```

```
ubuntu@ip-172-31-41-188:~$ ps aux --sort=-%mem | head -10
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         736  0.0  2.0 2025016 40292 ?       Ssl  05:45   0:00 /usr/lib/snapd/snapd
root         848  0.0  1.6 126208 32260 ?        Ssl  05:45   0:00 /usr/bin/python3 /usr/sha
e/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
root         729  0.0  1.5  46808 29892 ?        Ss   05:45   0:00 /usr/bin/python3 /usr/bin
networkd-dispatcher --run-startup-triggers
root        1133  0.0  1.0 1759024 21008 ?       Ssl  05:45   0:00 /snap/amazon-ssm-agent/13
09/amazon-ssm-agent
root           1  0.1  0.8  25212 16112 ?        Ss   05:45   0:02 /sbin/init
root         740  0.0  0.7 543288 14768 ?        Ssl  05:45   0:00 /usr/libexec/udisks2/udis
sd
systemd+     198  0.0  0.7  22716 14156 ?        Ss   05:45   0:00 /usr/lib/systemd/systemd-
esolved
root         129  0.0  0.7  50116 14108 ?        S<s  05:45   0:00 /usr/lib/systemd/systemd-
ournald
root         203  0.0  0.6  39048 13648 ?        Ss   05:45   0:00 /usr/lib/systemd/systemd-
devd
```

Disk and inode usage

```
ubuntu@ip-172-31-41-188:~$ df -h
Filesystem       Size  Used Avail Use% Mounted on
/dev/root         28G  2.1G   26G   8% /
tmpfs            953M     0  953M   0% /dev/shm
tmpfs            382M  892K  381M   1% /run
efivarfs         128K  3.1K  120K   3% /sys/firmware/efi/efivars
tmpfs            953M     0  953M   0% /tmp
none             1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
none             1.0M     0  1.0M   0% /run/credentials/systemd-resolved.service
/dev/nvme0n1p13  989M   96M  827M  11% /boot
/dev/nvme0n1p15  105M  6.3M   99M   7% /boot/efi
none             1.0M     0  1.0M   0% /run/credentials/systemd-networkd.service
none             1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
none             1.0M     0  1.0M   0% /run/credentials/serial-getty@ttyS0.service
tmpfs            191M  8.0K  191M   1% /run/user/1000
ubuntu@ip-172-31-41-188:~$ df -i
Filesystem       Inodes IUsed   IFree IUse% Mounted on
/dev/root       3637760 96387 3541373    3% /
tmpfs            243867     2  243865    1% /dev/shm
tmpfs            819200   744  818456    1% /run
efivarfs              0     0       0     - /sys/firmware/efi/efivars
tmpfs           1048576    16 1048560    1% /tmp
none               1024     1    1023    1% /run/credentials/systemd-journald.service
none               1024     1    1023    1% /run/credentials/systemd-resolved.service
/dev/nvme0n1p13   65536   647   64889    1% /boot
/dev/nvme0n1p15       0     0       0     - /boot/efi
none               1024     1    1023    1% /run/credentials/systemd-networkd.service
none               1024     1    1023    1% /run/credentials/getty@tty1.service
none               1024     1    1023    1% /run/credentials/serial-getty@ttyS0.service
tmpfs             48773    35   48738    1% /run/user/1000
ubuntu@ip-172-31-41-188:~$ lsblk
NAME         MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0          7:0    0   74M  1 loop /snap/core22/2411
loop1          7:1    0 28.2M  1 loop /snap/amazon-ssm-agent/13009
loop2          7:2    0 49.3M  1 loop /snap/snapd/26865
nvme0n1      259:0    0   30G  0 disk
├─nvme0n1p1  259:1    0 28.9G  0 part /
├─nvme0n1p13 259:2    0 1023M  0 part /boot
├─nvme0n1p14 259:3    0    4M  0 part
└─nvme0n1p15 259:4    0  106M  0 part /boot/efi
ubuntu@ip-172-31-41-188:~$ sudo du -xhd1 /var 2>/dev/null | sort -h
4.0K    /var/backups
4.0K    /var/crash
4.0K    /var/local
4.0K    /var/mail
4.0K    /var/opt
```

```
ubuntu@ip-172-31-41-188:~$ sudo du -xhd1 /var 2>/dev/null | sort -h
4.0K    /var/backups
4.0K    /var/crash
4.0K    /var/local
4.0K    /var/mail
4.0K    /var/opt
16K     /var/spool
40K     /var/snap
48K     /var/tmp
17M     /var/log
35M     /var/cache
207M    /var/lib
257M    /var
```

Application logs

```
ubuntu@ip-172-31-41-188:~/monitor-demo$ curl -I http://127.0.0.1:3000/
HTTP/1.0 200 OK
Server: SimpleHTTP/0.6 Python/3.14.4
Date: Tue, 20 Oct 2026 06:13:14 GMT
Content-type: text/html
Content-Length: 176
Last-Modified: Tue, 20 Oct 2026 06:12:02 GMT

ubuntu@ip-172-31-41-188:~/monitor-demo$ ps -p "$(cat ~/monitor-demo/app.pid)" -o pid,cmd
    PID CMD
   1599 python3 -m http.server 3000 --bind 127.0.0.1
ubuntu@ip-172-31-41-188:~/monitor-demo$ sudo ss -lntp | grep ':3000'
LISTEN 0      5          127.0.0.1:3000      0.0.0.0:*    users:(("python3",pid=1599,fd=3))

ubuntu@ip-172-31-41-188:~/monitor-demo$ cat ~/monitor-demo/app.log
127.0.0.1 - - [20/Oct/2026 06:13:14] "HEAD / HTTP/1.1" 200 -
ubuntu@ip-172-31-41-188:~/monitor-demo$ curl -sS -o /dev/null --max-time 10 \
  -w 'HTTP=%{http_code} total=%{time_total}s\n' \
  http://127.0.0.1:3000/
HTTP=200 total=0.000964s
ubuntu@ip-172-31-41-188:~/monitor-demo$ top -p "$(cat ~/monitor-demo/app.pid)"
top - 06:15:02 up 29 min,  1 user,  load average: 0.00, 0.00, 0.00
Tasks:   1 total,   0 running,   1 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   1905.2 total,   1318.6 free,    405.0 used,    347.5 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.   1500.2 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   1599 ubuntu    20   0  109180  22752  10576 S   0.0   1.2   0:00.08 python3
```